

The pizzetti proofs, explained

Part 2: The mean-value expansion and Appendix B

Companion to the pizzetti paper (Section 4.2 and Appendix B, including B.1)

Abstract

Part 1 grouped the series by powers of ε . Here the same series is regrouped by powers of the planet's size p . This connects it to a 1909 formula by Pizzetti for the average of a function over a disk, and to the small-planet and `qpower2` approximations used in practice. We then prove the two error bounds of Appendix B, going through the proof of Appendix B.1 line by line. The tools are Taylor expansion, the Laplacian, Gauss's divergence theorem in two dimensions (familiar from electrostatics), and one theorem about power series (Abel's). Every formula is checked in `validation/03_meanvalue_appendixB.ipynb`.

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1 The idea: average of a function over a small disk

Take any smooth function $f(x, y)$ and average it over a disk of radius p centred at a point c . If p is small, the average is close to $f(c)$. How close? Expand f around c in a Taylor series. Terms of odd order (like x , x^3 , xy^2) average to zero by symmetry: for every point of the disk the mirror point gives the opposite value. Terms of second order give $\langle x^2 \rangle = \langle y^2 \rangle = p^2/4$ and $\langle xy \rangle = 0$, so the second-order part of the Taylor series averages to

$$\frac{1}{2}f_{xx}\langle x^2 \rangle + \frac{1}{2}f_{yy}\langle y^2 \rangle = \frac{p^2}{8}(f_{xx} + f_{yy}) = \frac{p^2}{8}\Delta f(c),$$

with the Laplacian $\Delta f = f_{xx} + f_{yy}$. Continuing to all orders gives the *mean-value expansion*

$$\langle f \rangle = \sum_{n=0}^{\infty} \frac{(p^2/4)^n}{n!(n+1)!} \Delta^n f(c) = f(c) + \frac{p^2}{8}\Delta f(c) + \frac{p^4}{192}\Delta^2 f(c) + \dots \quad (1)$$

(Δ^n is the Laplacian applied n times). Only Laplacians appear because averaging over a disk does not prefer any direction, and the Laplacian is the only second-order operation without a preferred direction. The formula is the area version of a formula Pizzetti (1909) gave for averages over spheres; Section 3 proves it, with an exact remainder.

Worked example. For $f = x^2 + y^2$ and $c = 0$: the disk average of r^2 is $p^2/2$, and (1) gives $0 + \frac{p^2}{8} \cdot 4 = p^2/2$, since $\Delta(x^2 + y^2) = 4$ and all higher Laplacians vanish.

2 Applied to a star: the familiar approximations

For the star, $f = (1 - r^2)^\gamma$, the intensity term $\mu^{2\gamma}$, and c is the planet's centre at distance z . The first term of (1), $f(c) = (1 - z^2)^\gamma$, is the *small-planet approximation*: the brightness at the planet's centre times its area. The first two terms, $I + \frac{p^2}{8}\Delta I$, are exactly the qpower2 approximation of Maxted & Gill (2019) for the power-2 law, and the first-order Rossiter–McLaughlin formula of Ohta et al. (2005). So these known approximations are the beginning of one series, and the paper's polynomial is the same series, summed further.

Computing the Laplacians. For a function of r alone, the Laplacian in the plane is $f'' + f'/r$. For a power of $w = 1 - r^2$ this gives (paper, Eq. 22)

$$\Delta w^s = 4s(s-1)w^{s-2} - 4s^2w^{s-1}. \quad (2)$$

Derivation: $\frac{d}{dr}w^s = -2rs w^{s-1}$ and $\frac{d^2}{dr^2}w^s = 4r^2s(s-1)w^{s-2} - 2s w^{s-1}$; add $\frac{1}{r}\frac{d}{dr}w^s = -2s w^{s-1}$ and use $r^2 = 1 - w$. Applying (2) repeatedly produces every $\Delta^n f$ as a short sum of powers of w . For $\gamma = \frac{1}{2}$ the first four terms of (1) become the paper's Eq. (23):

$$D_{1/2} = \sqrt{w} \left[1 - \frac{p^2}{8}(v+v^2) + \frac{p^4}{192}(v^2+6v^3-15v^4) - \frac{p^6}{1024}(v^3+15v^4-105v^5+105v^6) + \dots \right], \quad v = \frac{1}{w}.$$

Why this is the same series as in Part 1. Part 1 wrote D_γ as a sum over powers of ε . Each term there is a polynomial in p^2 and z^2 . Collecting all terms with the same power p^{2n} gives a term T_n (paper, Eq. 20). Both series are power series in p^2 for the same quantity, and a power series has only one set of coefficients. So T_n *must* equal the n -th term of (1). (Regrouping is allowed because the series of Part 1 converges absolutely.)

Worked example. For $\gamma = \frac{1}{2}$, $p = 0.1$, $z = 0.5$: $T_0 = 0.866025$, $T_1 = -3.37 \times 10^{-3}$, $T_2 = -1.42 \times 10^{-5}$, $T_3 = -1.67 \times 10^{-7}$. All terms after the first are negative, like the terms of Part 1. (They are sums of Part 1's terms, which all have one sign for $0 < \gamma < 1$.)

Check it yourself. `validation/03_meanvalue.appendixB.ipynb`, cells 3a–3b: (2) and the $\gamma = \frac{1}{2}$ expansion are verified symbolically (sympy computes the Laplacians from scratch), and T_n agrees with the n -th mean-value term to 40 digits for several γ .

3 Appendix B.1, step by step: the mean-value expansion with remainder

We now prove (1) again, this time keeping track of what is left over when we stop after N terms. The argument has four steps.

Step 1: Circle averages and Gauss's law

Let $m_f(\rho)$ be the average of f over the *circle* of radius ρ around c . How does it change with ρ ? Its derivative is the average of the outward slope $\partial f/\partial \rho$ over the circle. The integral of the outward slope around the circle is the *flux* of the gradient ∇f through the circle, and by the divergence theorem (Gauss's law in two dimensions) that flux equals the integral of $\nabla \cdot \nabla f = \Delta f$ over the enclosed disk:

$$2\pi\rho m'_f(\rho) = \oint \frac{\partial f}{\partial \rho} d\ell = \int_{\text{disk of radius } \rho} \Delta f dA = 2\pi \int_0^\rho s m_{\Delta f}(s) ds.$$

In the last step the disk is cut into thin rings of radius s and circumference $2\pi s$, on each of which the average of Δf is $m_{\Delta f}(s)$. In words: *the circle average grows at a rate set by the average Laplacian inside it*, just as the electric field through a circle is set by the enclosed charge.

Step 2: Integrate once

Divide by $2\pi\rho$, integrate from 0 to ρ , and use $m_f(0) = f(c)$:

$$m_f(\rho) = f(c) + \int_0^\rho \frac{1}{r} \int_0^r s m_{\Delta f}(s) ds dr.$$

Swap the two integrals (the region is $0 \leq s \leq r \leq \rho$; for fixed s , r runs from s to ρ , and $\int_s^\rho dr/r = \ln(\rho/s)$):

$$m_f(\rho) = f(c) + \int_0^\rho s \ln \frac{\rho}{s} m_{\Delta f}(s) ds = f(c) + (\mathcal{T}m_{\Delta f})(\rho). \quad (3)$$

We call the integral operation $\mathcal{T}$. Two properties matter:

- its weight $s \ln(\rho/s)$ is *never negative* for $0 \leq s \leq \rho$;
- it maps powers to powers, $\mathcal{T}[\rho^{2j}] = \rho^{2j+2}/(2j+2)^2$; to see this, substitute $s = \rho u$ and use $\int_0^1 u^{2j+1} \ln(1/u) du = 1/(2j+2)^2$.

Step 3: Repeat

Equation (3) holds for any smooth function, so apply it to Δf inside the integral: $m_{\Delta f} = \Delta f(c) + \mathcal{T}m_{\Delta^2 f}$. Substituting, and using $\mathcal{T}[1] = \rho^2/4$,

$$m_f(\rho) = f(c) + \frac{\rho^2}{4} \Delta f(c) + \mathcal{T}^2 m_{\Delta^2 f}(\rho).$$

Repeating $N+1$ times gives

$$m_f(\rho) = \sum_{j=0}^N \frac{\rho^{2j}}{4^j (j!)^2} \Delta^j f(c) + \underbrace{\mathcal{T}^{N+1} m_{\Delta^{N+1} f}(\rho)}_{\text{remainder}}. \quad (4)$$

The coefficients follow from the power rule: $\mathcal{T}$ applied j times to 1 gives $\rho^{2j}/(2^2 \cdot 4^2 \cdots (2j)^2) = \rho^{2j}/(4^j (j!)^2)$.

Step 4: From circles to the disk, and the size of the remainder

The disk average is the circle averages weighted by circumference: $\langle f \rangle = \frac{2}{p^2} \int_0^p \rho m_f(\rho) d\rho$. Applied to the terms of (4), $\frac{2}{p^2} \int_0^p \rho \cdot \rho^{2j} d\rho = \frac{p^{2j}}{j+1}$, which turns $\frac{1}{4^j (j!)^2}$ into $\frac{1}{4^j j! (j+1)!}$: exactly the coefficients of (1).

The remainder is built from $m_{\Delta^{N+1} f}$ by repeated application of $\mathcal{T}$ (weights ≥ 0) and the final disk average (weights ≥ 0). So it is a *weighted average with non-negative weights* of the circle averages of $\Delta^{N+1} f$:

$$\text{remainder} = \int_0^p K(s) m_{\Delta^{N+1} f}(s) ds, \quad K(s) \geq 0.$$

Two consequences:

1. The total weight $\int K$ is what the remainder would be if $\Delta^{N+1} f$ were the constant 1. Then the next term of the series is the whole remainder, so $\int K = (p^2/4)^{N+1}/((N+1)!(N+2)!)$.
2. Each circle average lies between the smallest and largest value of $\Delta^{N+1} f$ on the disk, and so does any weighted average of them. Hence there is a point ξ of the disk with

$$\langle f \rangle - \sum_{n=0}^N \frac{(p^2/4)^n}{n!(n+1)!} \Delta^n f(c) = \frac{(p^2/4)^{N+1}}{(N+1)!(N+2)!} \Delta^{N+1} f(\xi). \quad (5)$$

This is the analogue of the Lagrange remainder of a Taylor series: the next term, evaluated at an unknown point instead of the centre.

4 Proposition 1: a two-sided bound

Proposition 1 (as in the paper). For $f = (1 - r^2)^\gamma$ with $0 < \gamma < 1$, all $\Delta^n f$ with $n \geq 1$ are negative and their size increases with r . The remainder of the mean-value expansion after N terms therefore lies between the right-hand side of (5) evaluated at $r = \max(0, z - p)$ and at $r = z + p$.

Why all Laplacians are negative. Write f as a power series in r^2 : $f = \sum_m \beta_m r^{2m}$ with the binomial coefficients β_m of Part 1, all negative for $m \geq 1$ when $0 < \gamma < 1$. In the plane, $\Delta r^k = k^2 r^{k-2}$, so $\Delta^n r^{2m} = 4^n (m! / (m - n)!)^2 r^{2(m-n)}$ (positive). Hence

$$\Delta^n f = \sum_{m \geq n} \beta_m 4^n \left(\frac{m!}{(m - n)!} \right)^2 r^{2(m-n)}$$

is a power series in r^2 with only negative coefficients (for $n \geq 1$). Such a function is negative, and its size grows with r .

Where on the disk. The unknown point ξ lies somewhere on the planet's disk. The smallest distance from the stellar centre on that disk is $z - p$ if the planet does not cover the centre ($z \geq p$), and 0 if it does ($z < p$); the largest is $z + p$. Since $|\Delta^{N+1} f|$ increases with r , the remainder lies between its values at these two radii.

A correction found by the checks. An earlier draft of the paper wrote $|z - p|$ for the smallest radius. For $z < p$ that is wrong: the disk then contains the stellar centre, where $r = 0$. At $z = 0$ the two ends of the old bound coincide, and the true remainder is not equal to that single value. The validation notebook found this, and the paper now has $\max(0, z - p)$.

Worked example. $\gamma = \frac{1}{2}$, $p = 0.1$, $z = 0.5$:

N	lower end	true remainder	upper end
0	-4.00×10^{-3}	-3.38×10^{-3}	-2.99×10^{-3}
1	-2.67×10^{-5}	-1.43×10^{-5}	-8.87×10^{-6}
2	-5.03×10^{-7}	-1.70×10^{-7}	-7.13×10^{-8}

And at $z = 0$, $N = 1$: the true remainder is -4.182×10^{-6} ; the corrected bound gives $[-4.359, -4.167] \times 10^{-6}$ (contains it); the old one gave the single value -4.359×10^{-6} (does not).

Limitation. The lower end uses the largest radius $z + p$, where $\Delta^{N+1} f$ is largest. Near the stellar edge it blows up, roughly like $(p/(1 - z - p))^{2N+2}$: for $N = 2$ the bound spans $[-5 \times 10^{-7}, -7 \times 10^{-8}]$ at $z = 0.5$ but $[-5.1, -1 \times 10^{-5}]$ at $z = 0.88$ ($p = 0.1$). Proposition 1 is therefore useful only away from the edge. Proposition 2 fixes this for the quadratic law.

5 Proposition 2: a bound valid up to the edge (Abel)

Proposition 2 (as in the paper). For $\gamma = \frac{1}{2}$, $t = p^2$, $t_1 = (1 - z)^2$ and $0 \leq p \leq 1 - z$,

$$0 \leq S_N^P(z, p) - D_{1/2}(z, p) \leq \left(\frac{p}{1 - z} \right)^{2N+2} \left[S_N^P(z, 1 - z) - D_{1/2}(z, 1 - z) \right],$$

where $S_N^P = \sum_{n \leq N} T_n$, and $D_{1/2}(z, 1 - z)$, the average over the planet just touching the edge from inside, has the elementary formula of the paper's Eq. (B4).

The idea: view $D_{1/2}$ at fixed z as a power series in $t = p^2$, $D_{1/2} = \sum_n d_n t^n$ (so $T_n = d_n t^n$). The largest planet that fits is the one touching the edge, $t = t_1$. If we knew the error at t_1 , we could scale it down to smaller planets. Four small steps:

1. *All d_n with $n \geq 1$ are negative* (the T_n are, see above). So the remainder $D - S_N^P = \sum_{n>N} d_n t^n$ is negative: $S_N^P \geq D$ for every $t < t_1$. That is the left inequality.
2. *The series still converges at the edge, $t = t_1$.* The partial sums at t_1 decrease as N grows (we keep adding negative terms). They cannot fall below $D(z, 1 - z)$: for $t < t_1$ we have $S_N^P \geq D$, and letting $t \rightarrow t_1$ keeps the inequality because both sides are continuous. A decreasing sequence that is bounded below converges.
3. *And it converges to the right value.* Abel's theorem says: if a power series converges at the end point of its interval, its sum there is the limit of its values inside. So $\sum_n d_n t_1^n = D(z, 1 - z)$, and the error at the edge is $\sum_{n>N} |d_n| t_1^n = S_N^P(z, 1 - z) - D(z, 1 - z)$.
4. *Scale down.* For $t < t_1$ and $n > N$, $t^n = t_1^n (t/t_1)^n \leq t_1^n (t/t_1)^{N+1}$. Summing, $\sum_{n>N} |d_n| t^n \leq (t/t_1)^{N+1} \sum_{n>N} |d_n| t_1^n$, and $(t/t_1)^{N+1} = (p/(1 - z))^{2N+2}$. That is the right inequality.

Worked example. $\gamma = \frac{1}{2}$, $p = 0.1$, $z = 0.5$: the true errors $S_N^P - D$ are 3.38×10^{-3} , 1.43×10^{-5} and 1.70×10^{-7} for $N = 0, 1, 2$; the Abel bounds are 3.94×10^{-3} , 2.30×10^{-5} and 3.54×10^{-7} . The tangent-disk formula gives $D_{1/2}(0.5, 0.5) = 0.767449091228817$, the same as the defining integral to all digits shown.

Check it yourself. `validation/03_meanvalue.appendixB.ipynb`, cells 3c–3e: the tangent-disk formula against the defining integral (agreement 10^{-36}); Proposition 1 with both the corrected and the old smallest radius (only the old one fails, and only for $z < p$); Proposition 2 with no violations.

6 Summary

Regrouping the series of Part 1 by powers of the planet size gives Pizzetti's mean-value expansion, whose first one and two terms are the small-planet and qpowers2 approximations. Appendix B.1 proves the expansion with an exact remainder using one physical idea, Gauss's law for the gradient of f , and the fact that averages with non-negative weights stay between the minimum and maximum. Proposition 1 turns this into a two-sided bound that works away from the stellar edge; Proposition 2 uses the sign of the coefficients and Abel's theorem to obtain a bound that works up to the edge. Neither is used by the pizzetti code, which relies on Theorem 2 (Part 1); they explain how the known approximations fit in and how accurate they are.

Next: Part 3, the phases in which the planet crosses the stellar edge (Sections 5.1 and 5.3 and Appendix C).